Backend Engineering Mini Projects Suite

This repository contains ten independent backend services. Each service runs independently via Docker and can later be orchestrated with docker-compose or Kubernetes. Every project has its own Dockerfile, README, requirements.txt, src/, tests/, and config/. Shared libraries should live under /shared.

URL Shortener

Folder: 01-url-shortener
Objective: Build a production-inspired backend service.
Core Features: Hashing, redirects, Redis cache, analytics.
Suggested Stack: FastAPI, PostgreSQL, Redis (where applicable), Docker, Pytest.
Deliverables: REST APIs, OpenAPI docs, unit tests, Dockerfile, CI workflow, structured logging, health endpoint.

Authentication Service

Folder: 02-auth-service
Objective: Build a production-inspired backend service.
Core Features: JWT, Refresh Tokens, OAuth, RBAC.
Suggested Stack: FastAPI, PostgreSQL, Redis (where applicable), Docker, Pytest.
Deliverables: REST APIs, OpenAPI docs, unit tests, Dockerfile, CI workflow, structured logging, health endpoint.

API Rate Limiter

Folder: 03-rate-limiter
Objective: Build a production-inspired backend service.
Core Features: Redis sliding window/token bucket.
Suggested Stack: FastAPI, PostgreSQL, Redis (where applicable), Docker, Pytest.
Deliverables: REST APIs, OpenAPI docs, unit tests, Dockerfile, CI workflow, structured logging, health endpoint.

Mini S3 Storage

Folder: 04-file-storage
Objective: Build a production-inspired backend service.
Core Features: Object storage, signed URLs.
Suggested Stack: FastAPI, PostgreSQL, Redis (where applicable), Docker, Pytest.
Deliverables: REST APIs, OpenAPI docs, unit tests, Dockerfile, CI workflow, structured logging, health endpoint.

Notification Service

Folder: 05-notification-service
Objective: Build a production-inspired backend service.
Core Features: RabbitMQ, retries, workers.
Suggested Stack: FastAPI, PostgreSQL, Redis (where applicable), Docker, Pytest.
Deliverables: REST APIs, OpenAPI docs, unit tests, Dockerfile, CI workflow, structured logging, health endpoint.

Job Queue

Folder: 06-job-queue
Objective: Build a production-inspired backend service.

Core Features: Task execution, scheduling.
Suggested Stack: FastAPI, PostgreSQL, Redis (where applicable), Docker, Pytest.
Deliverables: REST APIs, OpenAPI docs, unit tests, Dockerfile, CI workflow, structured logging, health endpoint.

API Gateway

Folder: 07-api-gateway
Objective: Build a production-inspired backend service.
Core Features: Reverse proxy, auth, rate limiting.
Suggested Stack: FastAPI, PostgreSQL, Redis (where applicable), Docker, Pytest.
Deliverables: REST APIs, OpenAPI docs, unit tests, Dockerfile, CI workflow, structured logging, health endpoint.

Log Aggregator

Folder: 08-log-aggregator
Objective: Build a production-inspired backend service.
Core Features: Centralized logging/search.
Suggested Stack: FastAPI, PostgreSQL, Redis (where applicable), Docker, Pytest.
Deliverables: REST APIs, OpenAPI docs, unit tests, Dockerfile, CI workflow, structured logging, health endpoint.

Feature Flag Service

Folder: 09-feature-flags
Objective: Build a production-inspired backend service.
Core Features: Rollouts, targeting.
Suggested Stack: FastAPI, PostgreSQL, Redis (where applicable), Docker, Pytest.
Deliverables: REST APIs, OpenAPI docs, unit tests, Dockerfile, CI workflow, structured logging, health endpoint.

Mini Redis

Folder: 10-mini-redis
Objective: Build a production-inspired backend service.
Core Features: In-memory KV store, TTL, pub/sub.
Suggested Stack: FastAPI, PostgreSQL, Redis (where applicable), Docker, Pytest.
Deliverables: REST APIs, OpenAPI docs, unit tests, Dockerfile, CI workflow, structured logging, health endpoint.